Unit 3: (Ch. 2) Review Answers

CHAPTER 2

# REVIEW

## KEY QUESTIONS AND TERMS

1. c DOK 1
2. d DOK 1
3. d DOK 1
4. the number of individuals of that species that a particular environment can support DOK 1
5. The population size and growth rate increase rapidly during Phase 1, its growth rate slows down during Phase 2, and its growth stops and growth rate drops to zero during Phase 3. DOK 2
6. The population growth rate is decreasing because more organisms are leaving than entering, so the population is getting smaller. DOK 2
7. the number of individuals in a given area DOK 1
8. The birds exhibit clumped distribution, probably to help them avoid predators. DOK 2
9. a DOK 2
10. c DOK 1
11. The carrying capacity would increase. DOK 2
12. Changes in environmental conditions such as temperature and rainfall can reduce carrying capacity to a point at which populations are wiped out. DOK 2
13. The predator-prey relationship affects density. When there is a lot of prey, the population of the predator increases. As it increases, the population of the prey decreases. As the population of the prey decreases, so too does the population of the predator. The cycle continues to repeat. The rates of increase and decrease are directly connected to the density of the populations, so it is a density-dependent limiting factor. DOK 3
14. Overcrowding can cause animals to fight, leading to possible disease or death; it can also cause females to neglect or kill young. These effects can reduce population growth by lowering birthrates or raising death rates—or both—and by increasing emigration. DOK 2
15. Sample answers: hunting, insect spraying (i.e., mosquito control), habitat destruction DOK 2
16. b DOK 1
17. c DOK 1
18. High birthrates and death rates shift to low birthrates and death rates. DOK 2
19. war, famine, disease DOK 1



CHAPTER 2

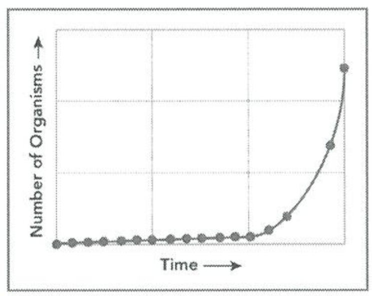
# REVIEW

## KEY QUESTIONS AND TERMS

### 2.1 How Populations Grow

HS-LS2-1, HS-LS2-2

1. The primary factors affecting population growth include birthrate, death rate, immigration, and
   a. density.
   b. geographic range.
   c. emigration.
   d. demography.
2. Which factor would reduce the likelihood of exponential growth occurring in a given population?
   a. lower death rates
   b. higher birthrates
   c. less competition
   d. reduced resources
3. The graph shows a model of the growth of a bacterial population. Which of the following correctly describes the growth curve?
   a. demographic
   b. logistic
   c. limiting
   d. exponential




4. What defines the carrying capacity of a particular species?
5. How does a population's growth rate change as it goes through the phases of logistic growth?
6. What is happening to the growth of a population in which death rate equals birthrate, but emigration is greater than immigration?
7. What is population density?
8. What type of population distribution pattern is exhibited by a flock of birds that nest close together? Why do they likely show this pattern?

### 2.2 Limits to Growth

HS-LS2-1, HS-LS2-2, HS-LS2-6, HS-LS4-5, EP&CIIIc

9. Which would be least likely to be affected by a density-dependent limiting factor?
   a. a small population of tigers scattered across a wide area
   b. a rapidly reproducing population of bacteria
   c. a large, dense population of fish in a pond
   d. a population of birds with a high immigration rate
10. Which of the following is a density-independent limiting factor?
    a. a struggle for food, water, space, or sunlight
    b. predator/prey relationships
    c. flood damage from a hurricane
    d. parasitism and disease
11. How might increasing the amount of a limiting nutrient in a pond affect the carrying capacity of the pond?
12. What are some limiting factors that might contribute to a species' extinction?
13. Why is the rise-and-fall cycle of a predator-prey relationship an example of a density-dependent limiting factor?
14. What are some consequences of stress from overcrowding, and how might they affect population growth?
15. What human activities are examples of density-independent limiting factors?

### 2.3 Human Population Growth

HS-LS2-1, HS-LS2-2, HS-ESS3-1

16. The scientific study of human populations is called
    a. ecology.
    b. demography.
    c. natural selection.
    d. transition.
17. Since the Industrial Revolution, human populations have
    a. decreased.
    b. reached carrying capacity.
    c. grown more rapidly.
    d. exceeded carrying capacity.
18. What shift in factors affecting population growth rates occurs during a country's demographic transition?
19. What were the limiting factors that Malthus identified as controls to human population increase?

1 Chapter 2 Populations

20. How do age-structure diagrams predict the growth of a population?

21. What are two ways in which the Irish potato famine led to a decrease in Ireland's population?

## CRITICAL THINKING

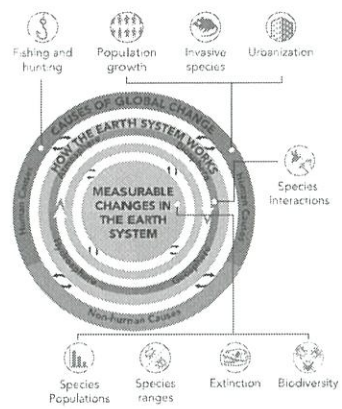
HS-LS2-1, HS-LS2-2, HS-LS2-6, HS-LS4-5, HS-ESS3-1

22. **Integrate Information** A scientist is studying a population of endangered Amargosa vole, a species native to California, to construct an outline of the effects on that population. What information about the vole population and the environment would the scientist want to include in the study?

23. **SEP Construct an Explanation** Asian carp were introduced to the lakes and rivers in the midwestern United States in the 1960s. Their U.S. population is increasing much more rapidly than native populations, and the species is considered invasive in the United States. Why is the U.S. population of carp increasing so rapidly? Incorporate the concepts of birthrate and death rate in your explanation.

24. **Reason Quantitatively** Two contained populations are both increasing in numbers even though their rates of growth are both decreasing. Population A is increasing faster than Population B. There is no immigration in either population. Explain the population increases and the difference between populations.

In this chapter, we have discussed how populations change over time. Use the diagram below to answer questions 25 and 26.




25. **Interpret Diagrams** Choose one human cause of global change. What "Earth System Processes" and "Measurable Changes" are connected to that human activity? Explain at least three connections.

26. **Evaluate** How has hunting changed sea otter populations? How might changes in sea otter populations affect the kelp forest ecosystem? Make connections among the Understanding Global Change topics in your explanation. (*Hint:* Look back at the diagrams in Chapter 1, as well as the diagrams in this chapter.)

27. **SEP Use Models** An aquarium is used to represent the population of fishes in a large lake. Describe one of the limiting factors that affects the fishes in the lake. Then, describe how the aquarium could model this factor.

28. Consider the differences between density-dependent and density-independent limiting factors.
    a. **Make a Claim** Determine which type of limiting factors, density-dependent or density-independent, have played the larger role in extreme environmental events, both historical and recent, that have affected human populations.
    b. **Use Evidence** Compile evidence from the chapter, including examples of extreme environmental events, to defend your claim.
    c. **Use Reasoning** Connect your claim to evidence to support your claim

29. a. **Apply Scientific Reasoning** Comparing age structure diagrams in **Figure 2-15**, predict the likely stages of the demographic transition that Japan and Niger are in. (You may wish to refer also to **Figure 2-14**.)
    b. **Predict** A researcher constructs a line graph illustrating predictions of future population growth in Niger, the United State, and Japan over the next five decades. What would the graphed curves most likely look like for each country? Include evidence to support your answes.



20. A higher percentage of young people indicates an increasing population and a higher percentage of old people indicates a decreasing population. **DOK 2**

21. Higher death rates due to starvation and increased emigration both led to a decrease in Ireland's population. **DOK 2**

## CRITICAL THINKING

22. The scientist would want to know the age structure of the vole population, as well as its geographic range (current and historical), growth rate, density, and distribution. The scientist would also want to know about resource availability and the effects of human activities. **DOK 3**

23. In the U.S., the Asian carp's birthrate is clearly much higher than its death rate, which is not the case where the species is native. In the U.S., the species must not have any natural predators to keep its population growth in check. **DOK 3**

24. Population A likely has a faster reproduction cycle, meaning more offspring can be born more quickly than in Population B. This would lead to a faster rate of growth in Population A. **DOK 3**

25. Sample answer: One human cause of global change is urbanization. Urbanization includes cutting down trees to build structures, highways, and agricultural fields. This affects the biosphere by decreasing both animal and plant biodiversity within the area. It also increases waste, which can pollute the hydrosphere. Also, by cutting down trees, there is less carbon dioxide removed from the atmosphere, and less oxygen released. **DOK 4**

26. Hunting caused a decline in the sea otter population by increasing the death rate. A decline in the sea otter population causes an increase in the sea urchin population since the sea urchin is the major food source for the sea otters. Sea urchins feed on kelp, so the the kelp population will subsequently decrease. In the Understanding Global Change model, this is reflected in species interactions, population sizes and ranges, and could cause extinction and loss of diversification in extreme cases. **DOK 3**

27. The amount of food in the lake is a limiting factor. The aquarium could model this factor by supplying the fish with only a certain amount of food per day or week. **DOK 3**

28. **a.** Density-independent factors such as Hurricane Maria hitting Puerto Rico, played a larger role in affecting the human population on Puerto Rico than density-dependent factors. **b.** A hurricane is a density-independent limiting factor. **c.** The population of Puerto Rico was hit with loss of life from the hurricane and the many month disruption in daily life will likely lead to fewer births and more deaths. The density-independent factor played more of a role than typical factors because the population of Puerto Rico was stable before the hurricane. **DOK 4**

29. **a.** Niger is in Stage 1 because a high number of young people means a high number of births and deaths. Japan is in stage 2 because it has a higher number of old people which indicates a decreasing population. **b.** The curve of Niger would be horizontal lines for both births and deaths which would be higher on the graph. The curve for the United States would be horizontal lines for both birth and death that would be lower on the graph. The curve for Japan would be rapidly decreasing. **DOK 4**